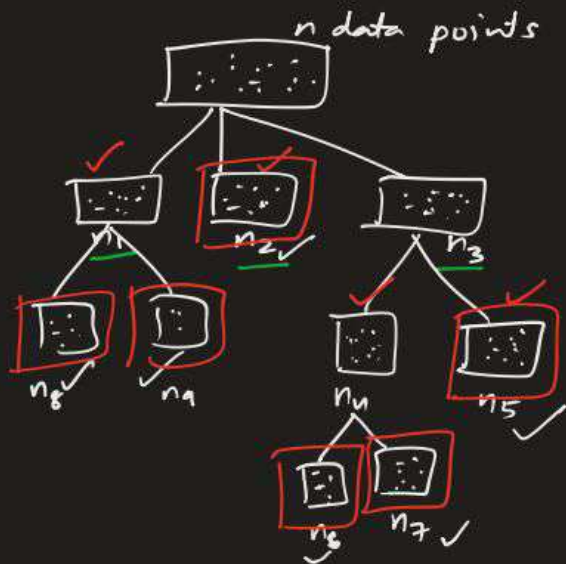


Divisive Hierarchical clustering (Top-Down H.C.)

clustering process starts with all observations in a single cluster and then recursively divides it into smaller clusters.

1. Consider all points in a single cluster
2. Split into clusters using any flat-clustering method, like k-means
3. Choose the best cluster among the clusters presents to split further, choose the one with largest sum of squares.
4. Repeat steps 2 & 3 until stopping criteria is reached
 - ↳ # clusters
 - ↳ within cluster sum of squares
 - ↳ single data point in every cluster.

0 - Average path in every cluster



Stop at 6 clusters

- ★ Computationally Complex

- * Flexible, optimally trying to minimize the variance.

$$=x=$$

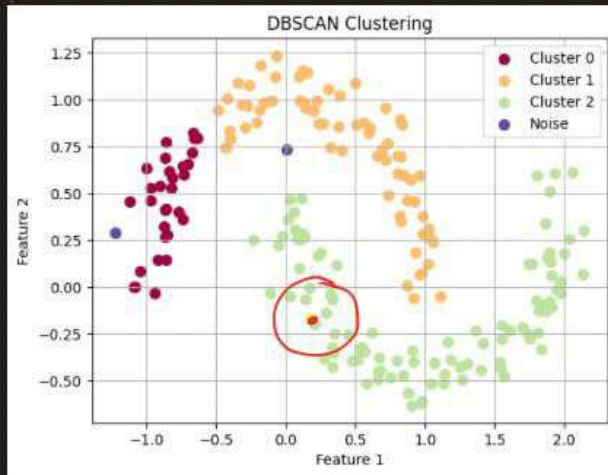
Density-Based Clustering

Groups together closely packed data points. Marks the outlier points as low-density regions. The algorithms can figure out any arbitrary shaped clusters.

DBSCAN Algorithm (Density-Based Spatial Clustering of Applications with Noise)

The algorithm works on two parameters

- epsilon (ϵ) : The maximum distance between two samples for one data point to be considered in the neighborhood of the other data point.
- minpts : The minimum number of points required to form a dense region.



1. Begin with arbitrary point and retrieve its neighbors within distance ϵ .
2. (i) If the number of neighbors is greater than minpts, mark this point as core point, otherwise as outlier

(ii) If the point is a core point, start a cluster from this point.

3. (i) For each core point, find all the neighbors which are also core points.

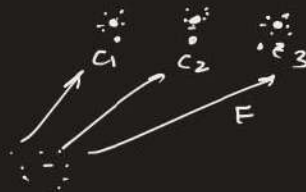
(ii) If they are, continue to add their neighbors to the cluster. This process continues until no other core points can be added to the cluster.

4. Repeat the process until we have no unvisited point left.

Distribution-Based Clustering

1. **Statistical Distributions:** In distribution-based clustering, each cluster is represented by a specific statistical distribution (e.g., Gaussian, Poisson). The goal is to model the distribution of each cluster in the dataset.
2. **Mixture Models:** Often, distribution-based clustering is implemented using mixture models, which assume that the data is generated from a mixture of several distributions. A common approach is the Gaussian Mixture Model (GMM), which assumes that the data points are generated from a mixture of several Gaussian distributions.
3. **Expectation-Maximization (EM) Algorithm:** To fit the parameters of the mixture model to the data, the EM algorithm is frequently used. It consists of two steps:
 - **Expectation Step (E-step):** Estimate the probability that each data point belongs to each cluster.
 - **Maximization Step (M-step):** Update the parameters of the distributions to maximize the likelihood of the data given the estimated probabilities.

K-means also uses EM algorithm



= x =